

孙宇石

手机 +86 18907174483 · 邮箱 ysunbp@gmail.com · 微信 sys18907174483

个人主页 <https://ysunbp.github.io> · GitHub <https://github.com/ysunbp> · Google Scholar [j9ypp3QAAAAJ](https://scholar.google.com/citations?user=j9ypp3QAAAAJ)

教育及工作经历

腾讯科技有限公司 · 互动娱乐事业群 光子工作室 · 高级研究员	2025.08 – 至今
研究方向: Data Agent、Agent Memory、强化学习 (AI Bot) 入选腾讯青云计划学者 (Project Up Scholar, 2025)	
蚂蚁研究院 · 研究实习生	2025.02 – 2025.05
方向: 多知识图谱增强的 RAG; 提交两项专利提案, 并以一作身份撰写论文	
香港科技大学 (HKUST) · 计算机科学与工程 · 博士	2021.09 – 2025.10
研究方向: 数据管护 (Data Curation)、大语言模型、检索增强生成 (RAG)、Agent Memory 导师: 陈雷教授 (IEEE Fellow, ACM Fellow) 获香港政府奖学金 (HKPFS) 资助	
香港科技大学 (HKUST) · 计算机科学与应用数学 · 本科	2017.09 – 2021.06
绩点 4.03/4.30 (Top 1% in HKUST) · 一等荣誉学位 曾赴新加坡南洋理工大学 (NTU)、瑞士苏黎世联邦理工学院 (ETH) 交换	

研究愿景

我的研究愿景是让 AI 系统能够可靠地基于数据进行推理。以数据管理为根基, 我构建面向 LLM 与 Agent 的检索增强生成 (RAG) 与智能体记忆 (Agent Memory), 把原始数据与知识转化为可信的模型行为——从问答到科学发现。

研究话题: RAG · Agent Memory · Knowledge Graphs · Text-to-SQL · Multimodal LLMs · AI for Science · Data Management

代表性论文与预印本 (* 共同一作, + 通讯作者)

预印本 · Preprints

Are LLMs Ready for Scientific Discovery? A Capability-Oriented Benchmark for AI Scientists
Chuhan Shi, Xiaoquan Ren, Sicheng Song, Haobo Li, Rui Sheng⁺, Yushi Sun⁺
Preprint 2026 [AI4S]

When Classic Cache Policies Fail: Learning-Augmented Replacement for Semantic Retrieval Buffers
Yushi Sun*, Bowen Cao*, Wai Lam
Preprint 2026 [RAG, Agent Memory]

STALE: Can LLM Agents Know When Their Memories Are No Longer Valid?
Hanxiang Chao*, Yihan Bai*, Rui Sheng, Tianle Li, Yushi Sun⁺
Preprint 2026 [Agent Memory]

GRAVITY: Architecture-Agnostic Structured Anchoring for Long-Horizon Conversational Memory
Yushi Sun*, Bowen Cao*, Dong Fang⁺, Lingfeng Su, Wai Lam
Preprint 2026 [Agent Memory]

LakeHopper: Cross Data Lakes Column Type Annotation through Model Adaptation
Yushi Sun, Xijia Li, Nan Tang, Quanqing Xu, Chuanhui Yang, Lei Chen
Preprint 2026 [Data Management]

MedKGI: Iterative Differential Diagnosis with Medical Knowledge Graphs and Information-Guided Inquiring

Qipeng Wang, Rui Sheng, Yafei Li, Huamin Qu, Yushi Sun⁺, Min Zhu⁺
Preprint 2026 [Knowledge Graphs, AI4S]

CacheRAG: A Semantic Caching System for Retrieval-Augmented Generation in KGQA

Yushi Sun, Lei Chen
Preprint 2026 [RAG, Knowledge Graphs]

已发表 · Published

VeriLLMed: Interactive Visual Debugging of Medical Large Language Models with Knowledge Graphs

Yurui Xiang, Xingyi Mao, Rui Sheng, Zixin Chen, Zelin Zang, Yuyang Wu, Haipeng Zeng, Huamin Qu, Yushi Sun⁺, Yanna Lin⁺
IEEE VIS 2026 [Knowledge Graphs]

Navigating the Mirage: A Dual-Path Agentic Framework for Robust Misleading Chart QA

Yanjie Zhang, Yafei Li, Rui Sheng, Zixin Chen, Yanna Lin, Huamin Qu, Lei Chen, Yushi Sun⁺
ACM MM 2026 (CCF-A) [Multimodal LLMs]

APEX-SQL: Talking to the Data via Agentic Exploration for Text-to-SQL

Bowen Cao, Weibin Liao, Yushi Sun⁺, Dong Fang⁺, Haitao Li, Wai Lam
ACM SIGKDD 2026 (CCF-A) [Text-to-SQL, Data Management]

KERAG: Knowledge-Enhanced Retrieval-Augmented Generation for Advanced Question Answering

Yushi Sun, Kai Sun, Ethan Yifan Xu, Xiao Yang, Xin Luna Dong, Nan Tang, Lei Chen
EMNLP 2025 (CCF-B) [RAG, Knowledge Graphs]

CRAG — Comprehensive RAG Benchmark

Xiao Yang*, Kai Sun*, Hao Xin*, Yushi Sun*, et al. (with Meta Reality Labs)
NeurIPS 2024 (CCF-A) [RAG]

Are Large Language Models a Good Replacement of Taxonomies?

Yushi Sun, Hao Xin, Kai Sun, Ethan Yifan Xu, Xiao Yang, Xin Luna Dong, Nan Tang, Lei Chen
VLDB 2024 (CCF-A) [Knowledge Graphs]

Cross-domain-aware Worker Selection with Training for Crowdsourced Annotation

Yushi Sun, Jiachuan Wang, Peng Cheng, Libin Zheng, Lei Chen, Jian Yin
ICDE 2024 (CCF-A) [Data Management]

RECA: Related Tables Enhanced Column Semantic Type Annotation Framework

Yushi Sun, Hao Xin, Lei Chen
VLDB 2023 (CCF-A) [Data Management]

在校经历与其它活动

共同主办 KDD Cup 2024 · 2024.01 – 2024.08

与 Meta Reality Labs 团队共同主办 KDD Cup 2024 (CRAG 赛道)；负责比赛数据准备，以共同一作身份在 NeurIPS 2024 发表 CRAG - Comprehensive RAG Benchmark。

教学助教 (TA)：COMP 1021 计算机科学导论 (2024 秋) · COMP 2711H 荣誉离散数学 (2022 秋) · COMP 5712 组合优化导论 (2022 春)

学术服务：

- 程序委员会 (PC Member)：AAAI 2027
- 会议审稿人 (Conference Reviewer)：ACM SIGKDD 2026 / 2027, ACM MM 2026, CIKM 2023
- 期刊审稿人 (Journal Reviewer)：TKDE 2024

荣誉奖项

2025

腾讯青云计划学者 (Tencent Project Up Scholar)

2023 / 2022 / 2021	红鸟学术卓越奖 (RedBird Academic Excellence Award)
2021	香港政府博士研究生奖学金 (HKPFS)
2021	HKUST Academic Achievement Medal (Top 1% in HKUST)
2020	HKSAR Government Scholarship Fund – Reaching Out Award
2020	Lee Hysan Foundation Scholarship
2019 / 2018	University's Scholarship Scheme for Continuing Undergraduate Students
2019	The Cheng Foundation Scholarships for Mainland Students
2017	Admission Scholarship (HKUST 优秀新生入学奖学金)